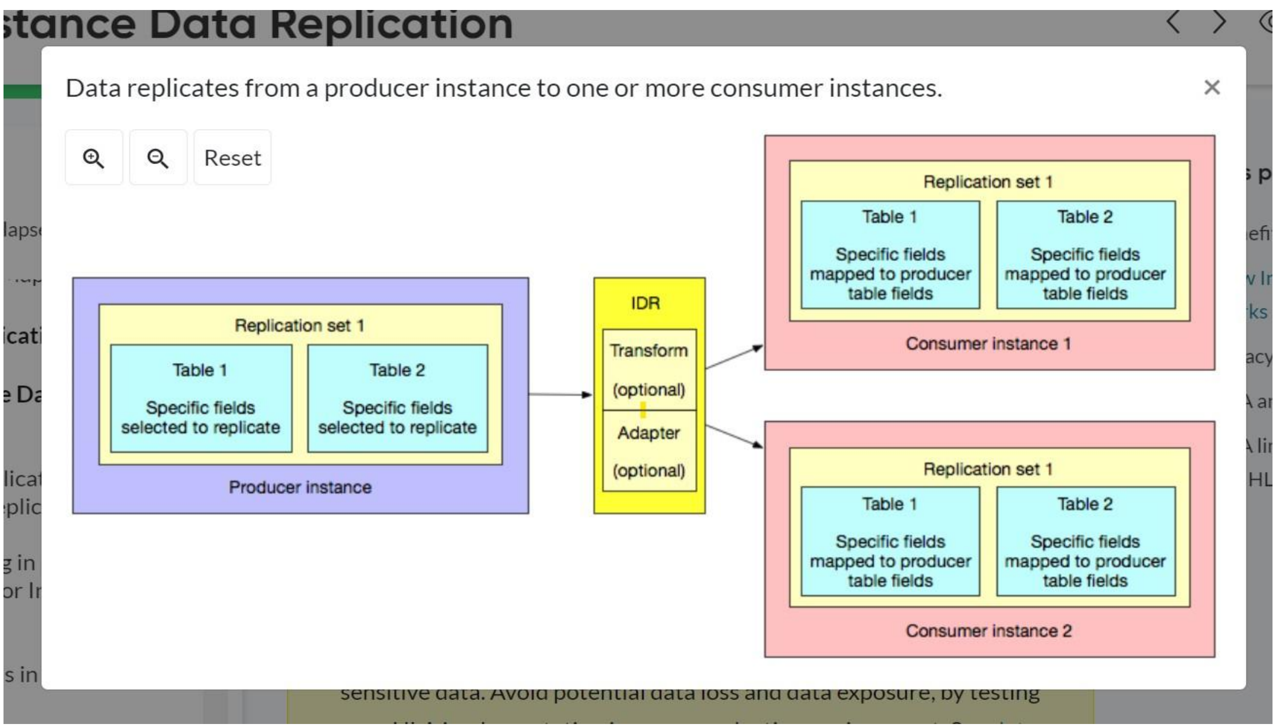
Instance Data Replication (IDR)

ServiceNow Instance Data Replication (IDR) copies data updates from one instance, called the **producer** instance, to one or more other instances called the **consumer** instances.

IDR enables you to maintain consistent data across different instances. For example, you can synchronize data between different organizations in your company or even between different companies with separate instances.

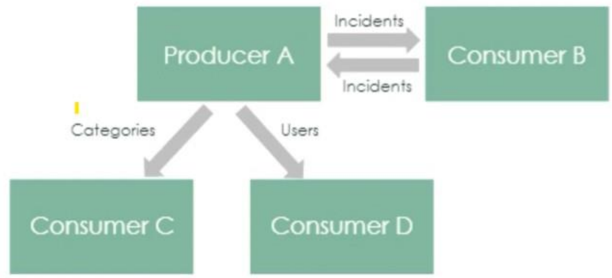
1. **One-to-Many Replication**: Propagates data from a **producer** instance to multiple **consumer** instances.



2. **Bidirectional Replication**: Allows data to flow back and forth between producer and consumer instances.

Many use cases for IDR

- Single source of truth for foundation data
- Manage data across instances
  - Separate instances for different departments
  - Dev/Production stack
  - External and Internal Instances
  - Managed Services
  - Merger between customers



Benefits

- Data is automatically replicated to one or more other instances.
- Data can be modified and mapped to any table and table column on other instances. For example, you can modify and map table columns to localize data for different locales.
- Data that is updated on consumer instances can be replicated to the producer instance.
- Data, such as problem requests, can be copied to consumer instances for third parties to use. The third party can update the problem issue on the consumer instance. The data can then be updated on the producer instance.
- Business rules can trigger post-replication workflows, such as generating notifications or validating the replication.
- Data that is in transit during a crash is recoverable.

Pre-requisites

The Instance Data Replication (IDR) plugin requires a separate subscription and must be activated by ServiceNow personnel.

Roles required: idr_admin or admin

Note: Detailed information about roles for IDR can be found here:
https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/reference/instance-data-replication-roles.html

The IDR plugin ID is **com.glide.idr**. (Beginning with the New York release, IDR is supported)

When purchasing an IDR subscription, ServiceNow, Inc. personnel also activate the following plugins that IDR depends upon:

1. **com.snc.db.data_replicate**
2. **com.glide.transform**
3. **com.glide.kmf**

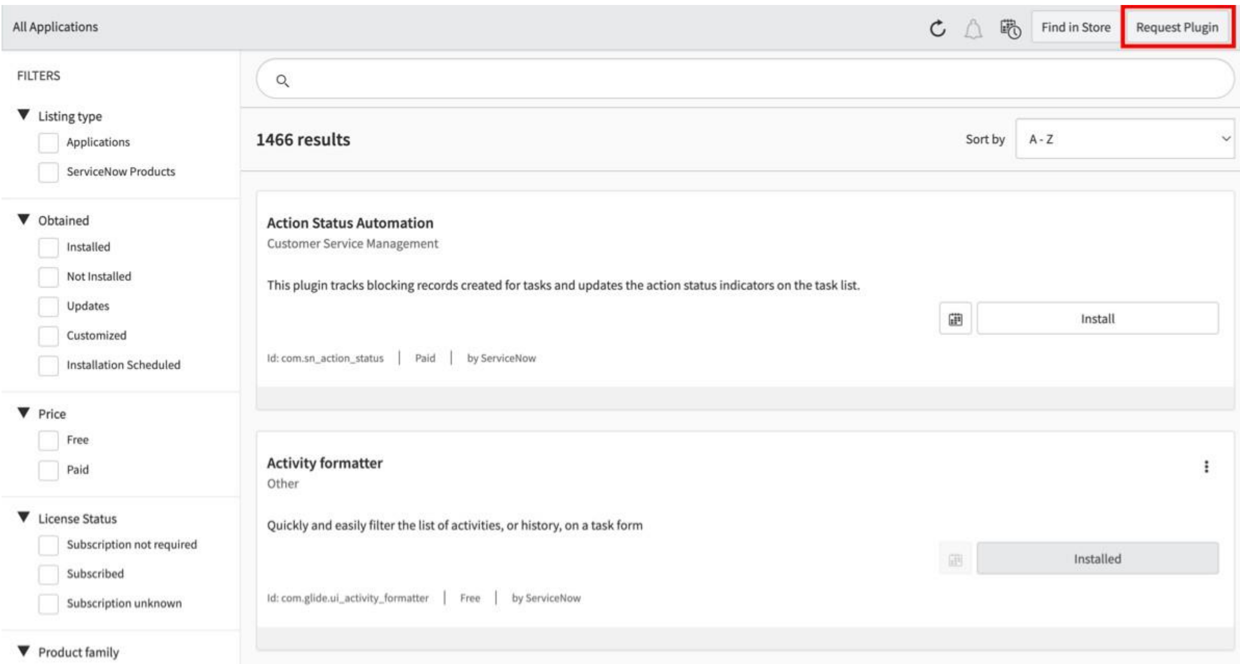
Note: If you want to evaluate the product on a subproduction instance without charge, follow these steps.

URL: https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/request-instance-data-replication.html

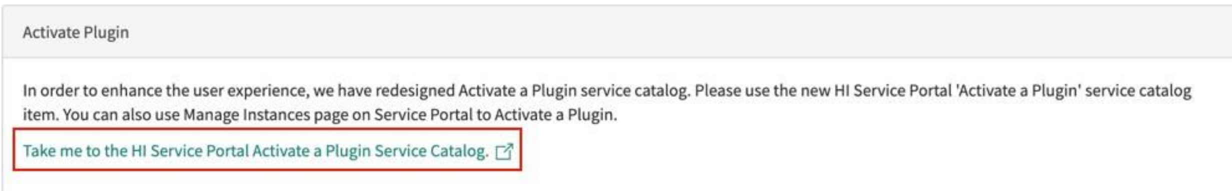
Procedure

1. Navigate to All > System Applications > All Available Applications > All.

2. On the All Applications page, select Request Plugin to open the Activate Plugin form on Now Support.



3. On Now Support, select the link to access the Now Support Service Portal Service Catalog.



4. Select your instance.

5. Select Actions > Activate Plugin.

6. On the Activate Plugin form, provide the following information:

Field	Description
What is your target instance	**Instance on which to activate the plugin.**
Which plugin would you like to activate	**Name of the plugin to activate.** **Note: If the system does not list the plugin you want or if you are activating the plugin on an OEM or on-premise instance, select the Plugin I'm looking for is not listed check box and then enter the name of the plugin.**
Select Maintenance Date and Time	**The date and time to activate the plugin.** **Note: Plugins are activated in two batches, once in the morning and once in the evening, on every business day in the US Pacific time zone. If the plugin must be activated at a specific time, enter the request in the Reason/Comments field.**

Preparing for instance data replication

Analyzing table hierarchies

We need to determine if the tables that needs to be replicated is a part of a parent-child table hierarchy.

If the tables are a part of a parent-child table hierarchy, we need to decide whether to replicate the data from the:

1. parent perspective (retaining only columns belonging to the parent table)

   or

2. from the child perspective (retaining all columns that belong to the child tables)

Note: To maintain data integrity and ensure that reference fields are populated on the consumer as you'd expect, you must replicate all of the parent and child tables in the table hierarchy.

Detailed information for preserving table hierarchy for IDR can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/preserving-table-hierarchy.html

Organizing replication sets

Strategy 1

Create a single replication set for each table.

This option requires more setup, configuration, and time management, but allows you to pause replication for a single table without impacting other tables.

- Replication is performed in parallel using multiple jobs.
- Throughput metrics are easier to track with separate replication sets.
- When an error occurs, replication for a single table is affected instead of all multiple tables.

Strategy 2

Create a replication set for a group of related tables.

This option is easier to set up and manage, but oversight and potential issues impact all tables in the group.

- If you need find information about replication status, the logical name you use for this replication set might be easier to find and understand compared to using distinct replication sets for each table.
- If you want to pause replication, you can do so for the entire group of tables by pausing replication from one replication set.
- If a replication issue occurs on one table and replication stops, this stops replication for all tables in the replication set.

Note: If we decide to group tables in a single replication set, try to limit it to five or fewer tables. If we have more than five tables to replicate, use multiple replication sets with five or fewer tables in each set.

Deciding which columns to replicate

Avoid including every column in your replication set by default.

For example, if you include a column updated by the system automatically on a frequent basis, IDR may replicate data more often than necessary. When the system replicates this data, it can negatively impact the capacity subscription.

Preparing target tables on the consumer instance

By default, IDR uses a record's sys_id field as the lookup value to keep data synchronized between the producer and consumer instance.

If the target table contains existing data or records from prior data imports, the sys_id values in the consumer don't match the sys_ids from the producer instance.

For optimal replication results, follow these guidelines:

- Make the producer instance the exclusive data source for the consumer instance.
- Prior to replication, ensure that the target table in the consumer instance is empty. Ideally, the initial records in the target instance are created from data sent exclusively from the producer via a replication set or a clone.
- Ensure that there are no other ongoing data imports or updates to target tables on the consumer instance after replication starts.

Note: Alternatively, we can use a custom Coalesce field to identify unique records (instead of the default sys_id column) for replication.

Use a custom Coalesce column when records on the consumer instance have a differentsys_id for the same records on the producer instance:

https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/concept/idr-coalesce.html

Reviewing business rules

1. We can use business rules to trigger workflows after replication, such as sending a notification or validating the replicated data.

https://docs.servicenow.com/bundle/xanadu-servicenow-platform/page/administer/instance-data-replication/task/post-replication- actions.html

2. Review any business rules on the consumer instance that might prevent the idr.system user from inserting, updating, or deleting records in the target table.

Reviewing ACLs

Review the ACLs in place on the target table in the consumer instance to ensure that IDR can replicate data successfully. Confirm that the **idr.system** user has the appropriate roles for the target table.

Reviewing data policies

Review any data policies in place on the target table in the consumer instance to ensure that IDR can replicate data successfully. Confirm that the data you are replicating satisfies the data policy.

Note: If failures related to **Business Rules, ACLs** and **Data Policies** occur, they appear in the **Instance Data Replication > Replication Payload Error** table on the consumer instance.

View the Error Message field for details on the data policy that is causing the failure.

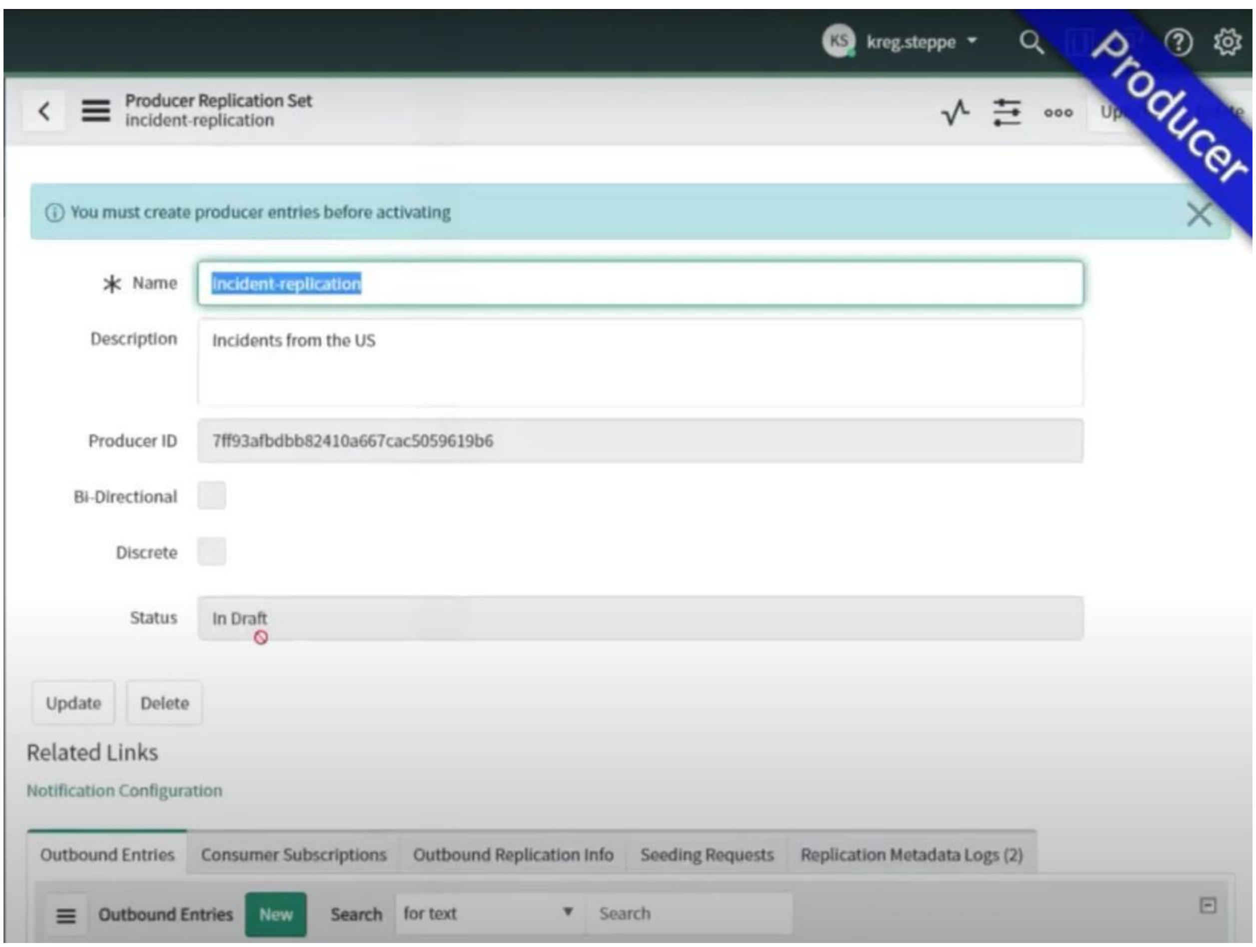
How Instance Data Replication works

We use the **Instance Data Replication plugin (com.glide.idr)** to replicate data updates on one instance, called the **producer** instance, to one or more other instances, called the **consumer** instances.

By configuring a **producer** replication set, we can specify the tables and table columns on the **producer** instance to replicate.

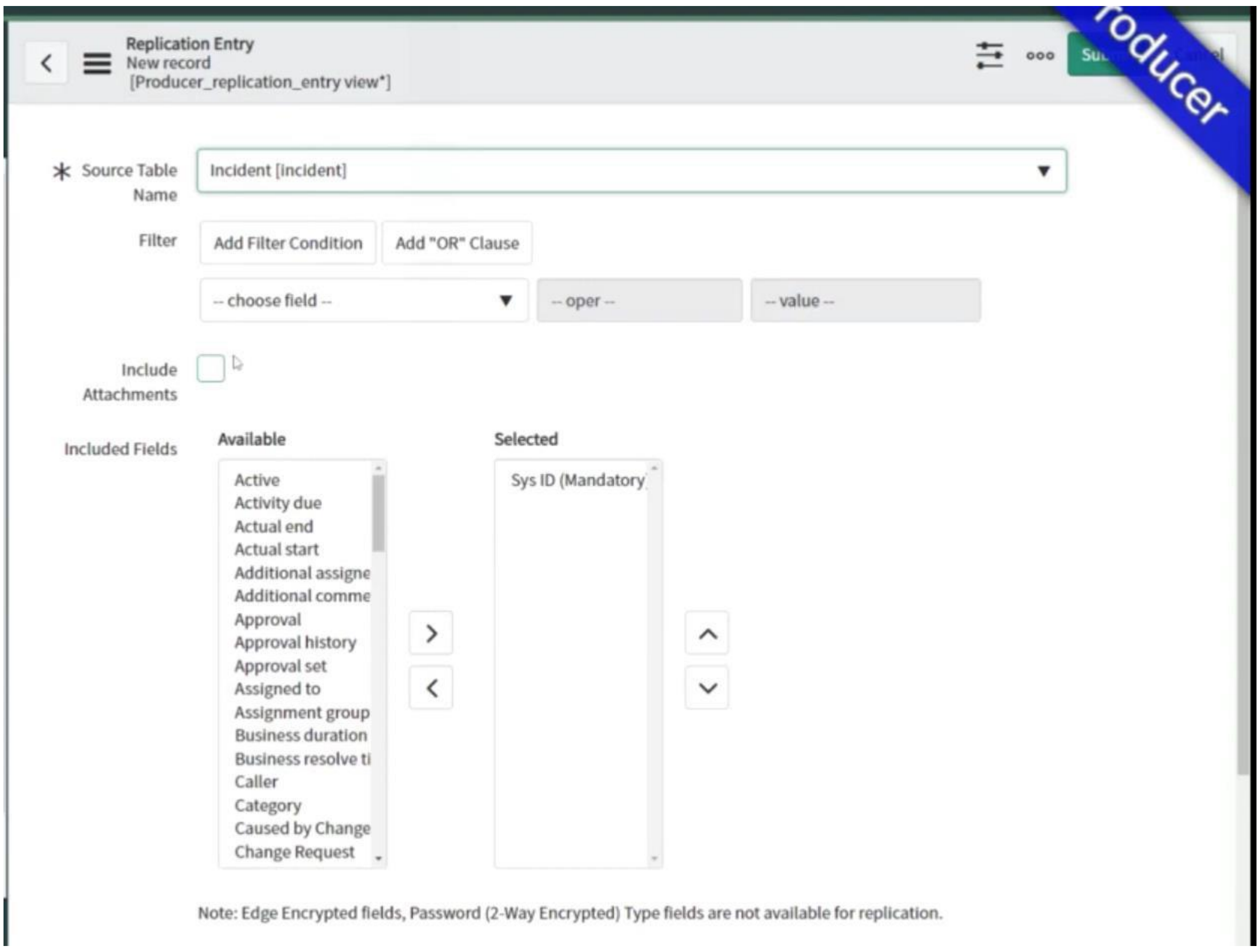
Note: Detailed information about setting up a Producer Replication link can be found here: https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/create-p-replication-set.html

1. Sample **Producer Replication Set** with status in **In Draft**

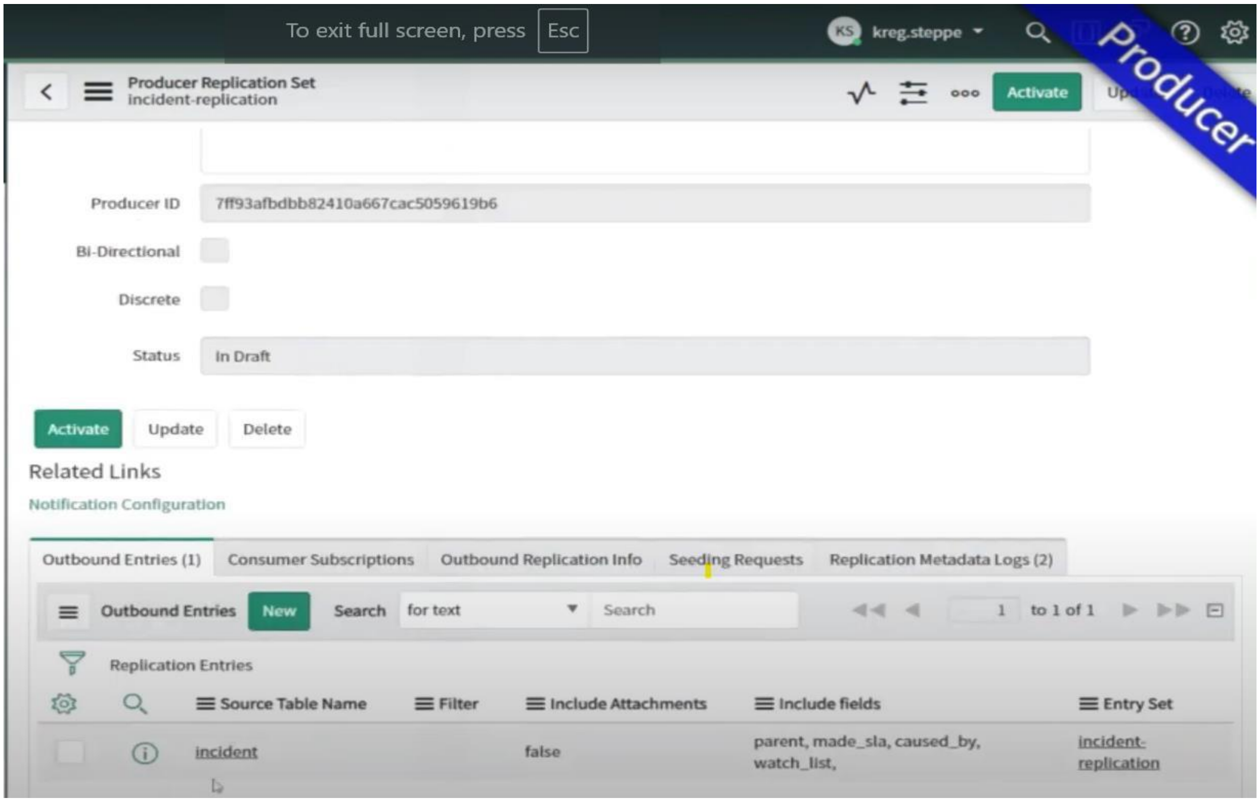


2. Cick on **New** option under **Outbound Entries** to select the table that needs to be exported. Here we have the option to select the table and :

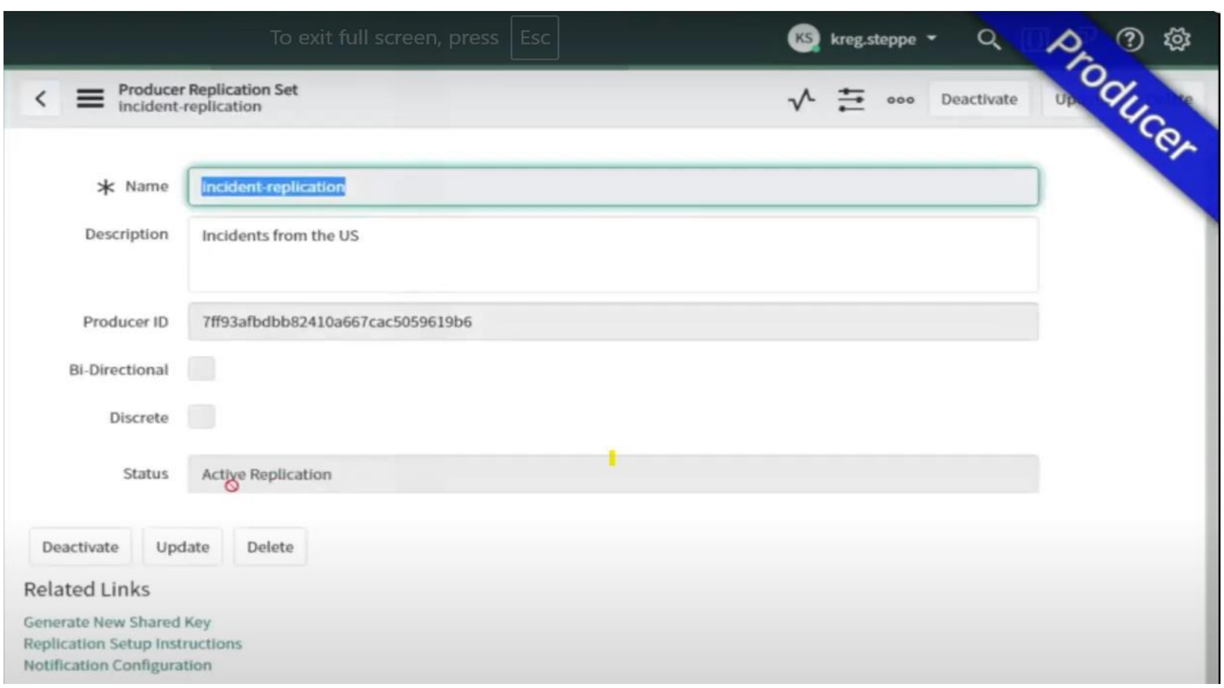
- set Filter conditions
- Option to include attachments
- Include Fields



3. Once an **Outbound Entries** is created, we have the option to **activate** the **Producer Replication Set**

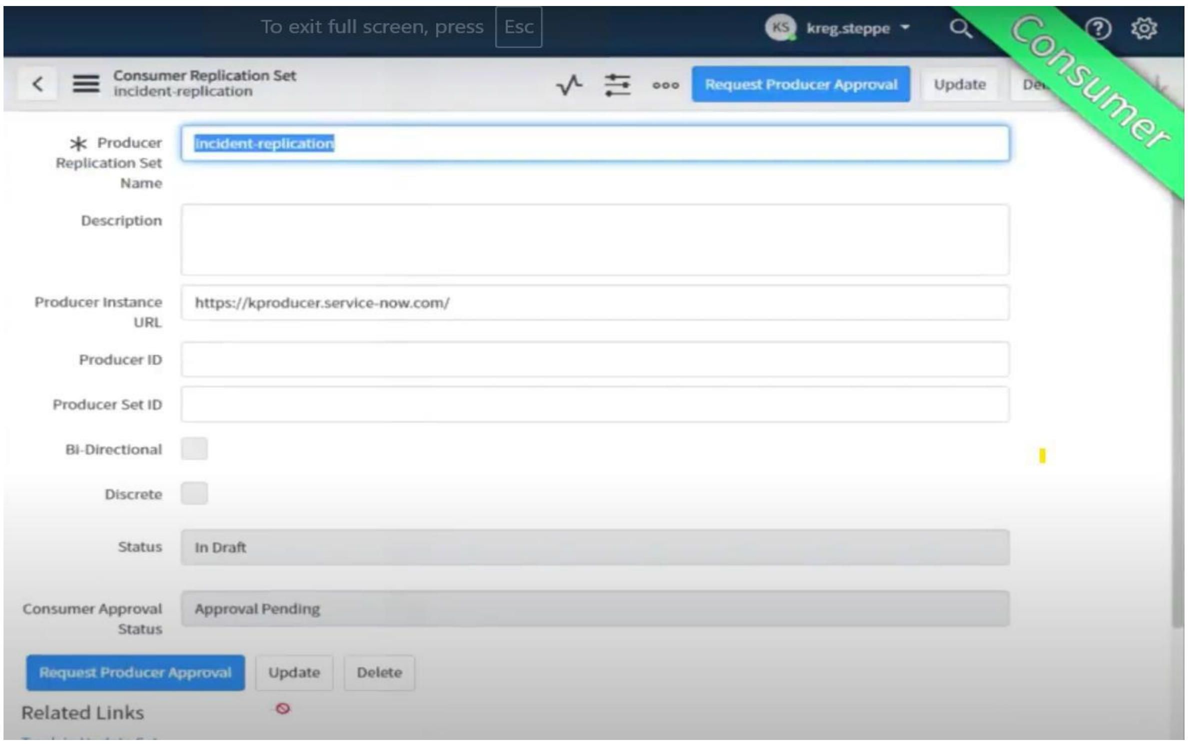


4. Producer Replication Set **in** Active Replication **status**



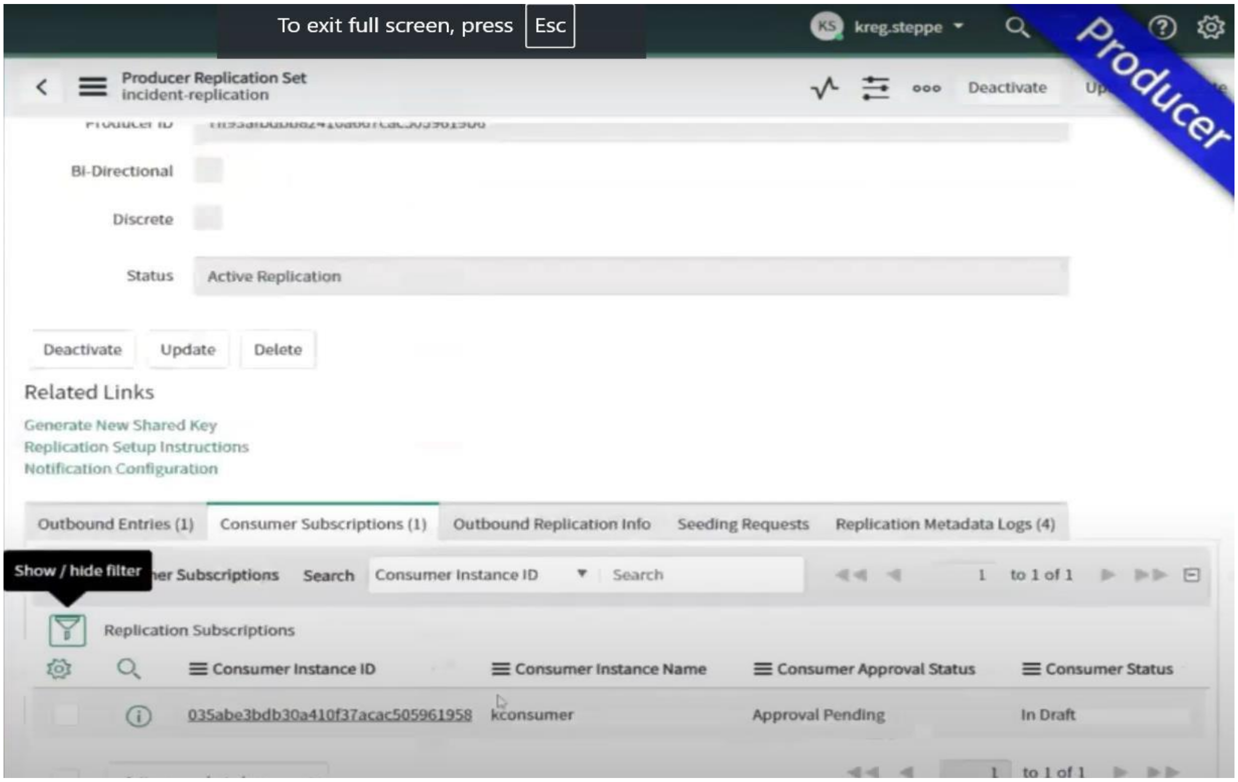
5. Configuring a sample **Consumer Replication Set** in Consumer instance, once created the status will be **Approval Pending**

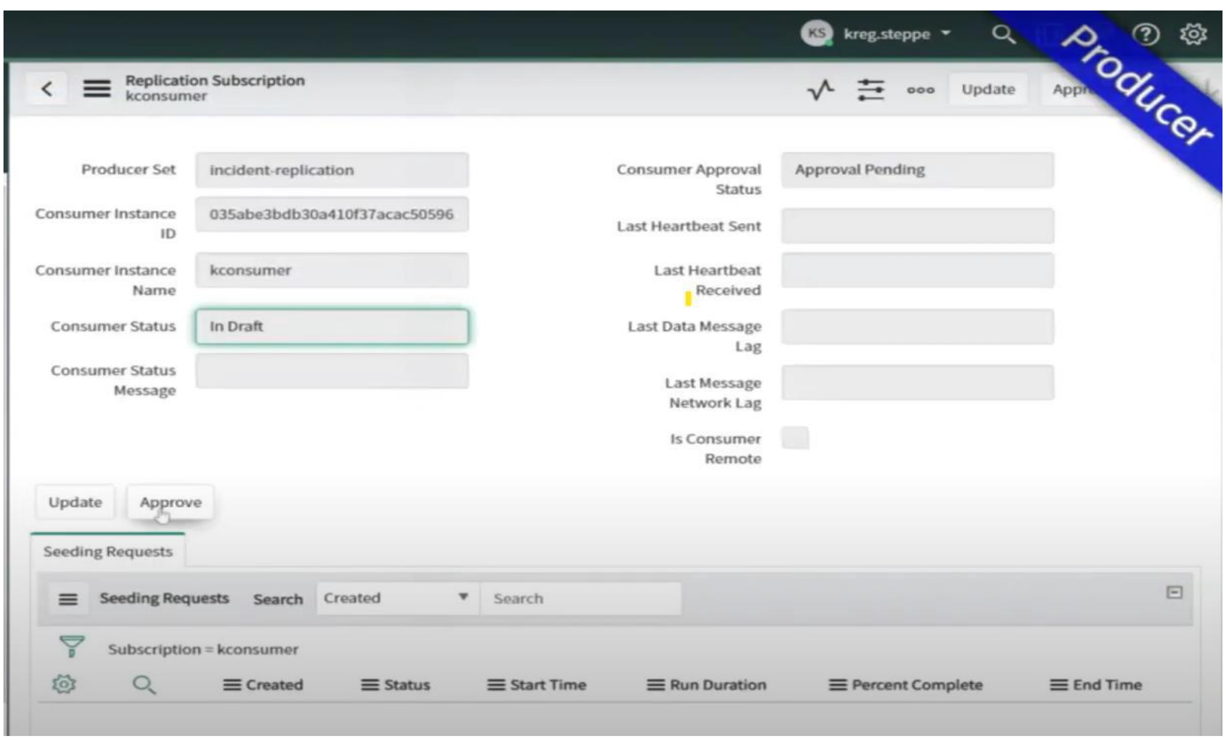
Detailed information about creating a **Consumer Replication Set** can be found here: https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/create-c-replication-set.html



6. Then go back to the **Producer Replication Set** > **Consumer Subsctiptions** to approve the request.

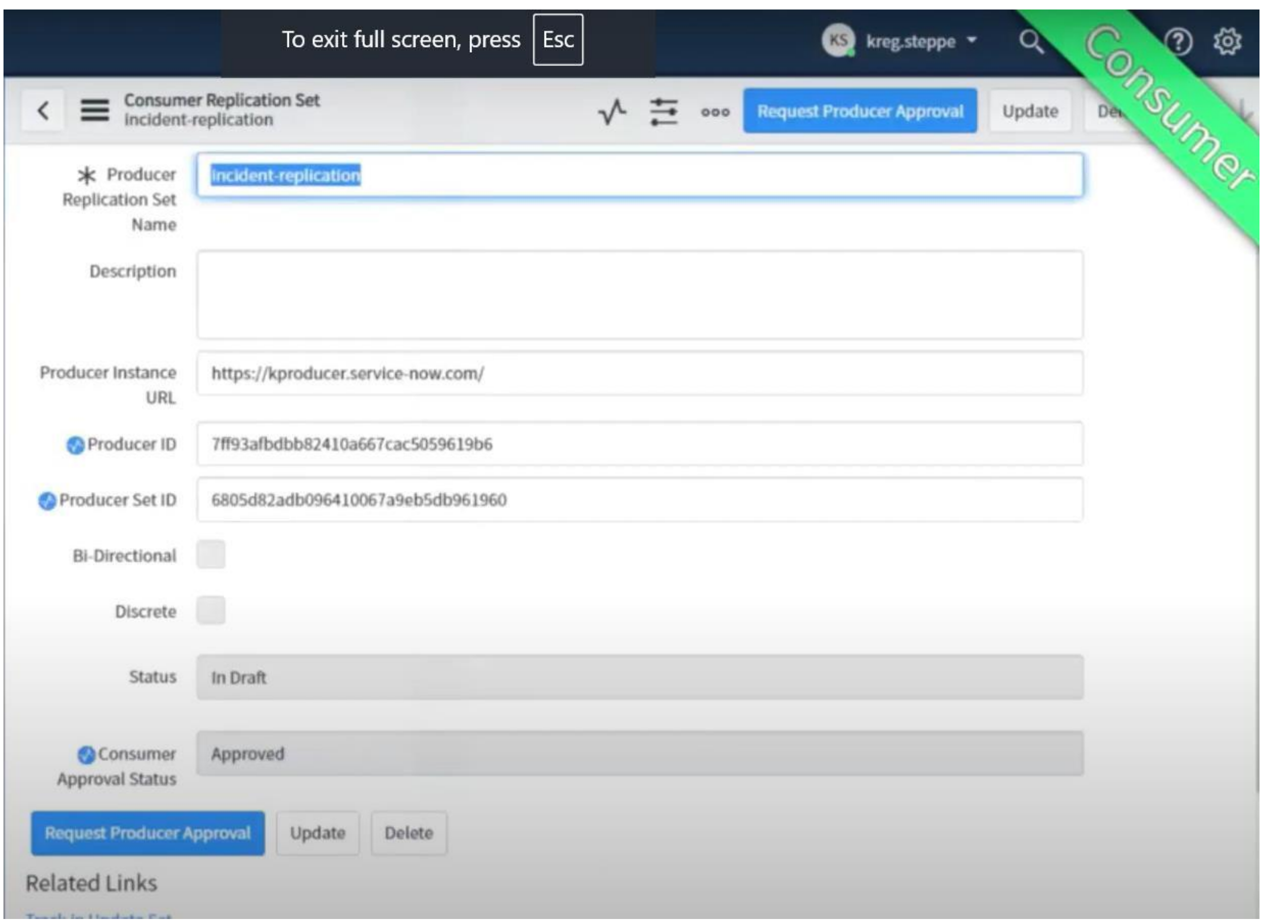
Detailed information about granting access to replicated data for IDR can be found here: https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/approve-consumer.html


To exit full screen, press Esc
kreg.steppe
Producer
Producer Replication Set
incident-replication
Deactivate
Bi-Directional
Discrete
Status
Active Replication
Deactivate
Update
Delete
Related Links
Generate New Shared Key
Replication Setup Instructions
Notification Configuration
Outbound Entries (1)
Consumer Subscriptions (1)
Outbound Replication Info
Seeding Requests
Replication Metadata Logs (4)
Show / hide filter
Search
Consumer Instance ID
Search
to 1 of 1
Replication Subscriptions
Consumer Instance ID
Consumer Instance Name
Consumer Approval Status
Consumer Status
035abe3bdb30a410f37acac505961958
kconsumer
Approval Pending
In Draft

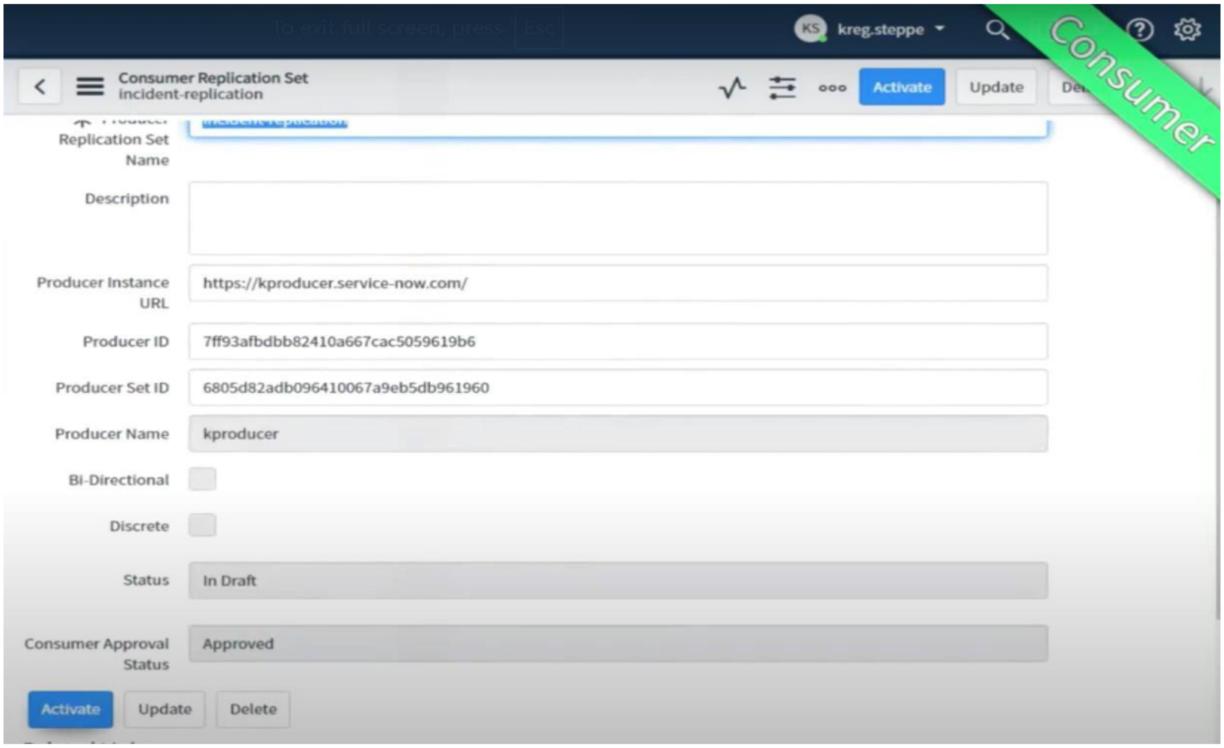

kreg.steppe
Producer
Replication Subscription
kconsumer
Update
Producer Set
incident-replication
Consumer Instance ID
035abe3bdb30a410f37acac50596
Consumer Instance Name
kconsumer
Consumer Status
In Draft
Consumer Status Message
Consumer Approval Status
Approval Pending
Last Heartbeat Sent
Last Heartbeat Received
Last Data Message Lag
Last Message Network Lag
Is Consumer Remote
Update
Approve
Seeding Requests
Seeding Requests
Search
Created
Search
Subscription = kconsumer
Created
Status
Start Time
Run Duration
Percent Complete
End Time

7. Click on **YES**

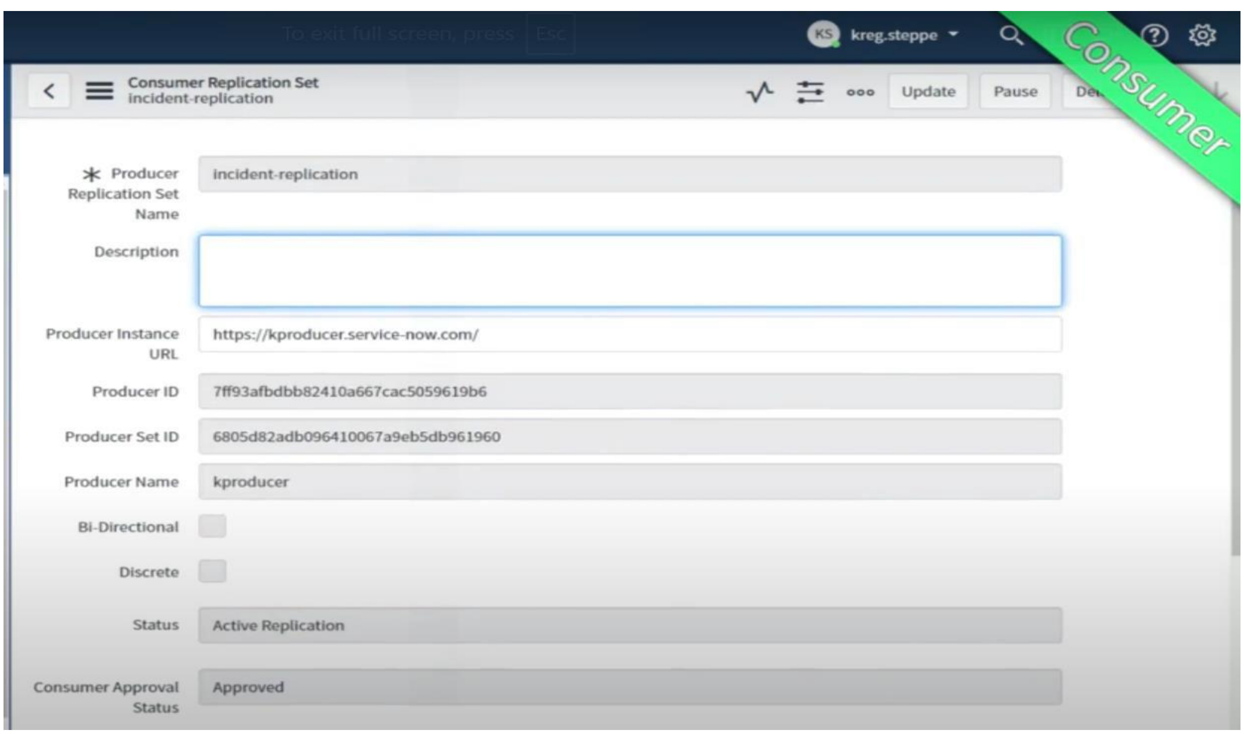
Once approved, the **Producer ID** and **Producer Set ID** fields will be updated in **Consumer Replication Set**



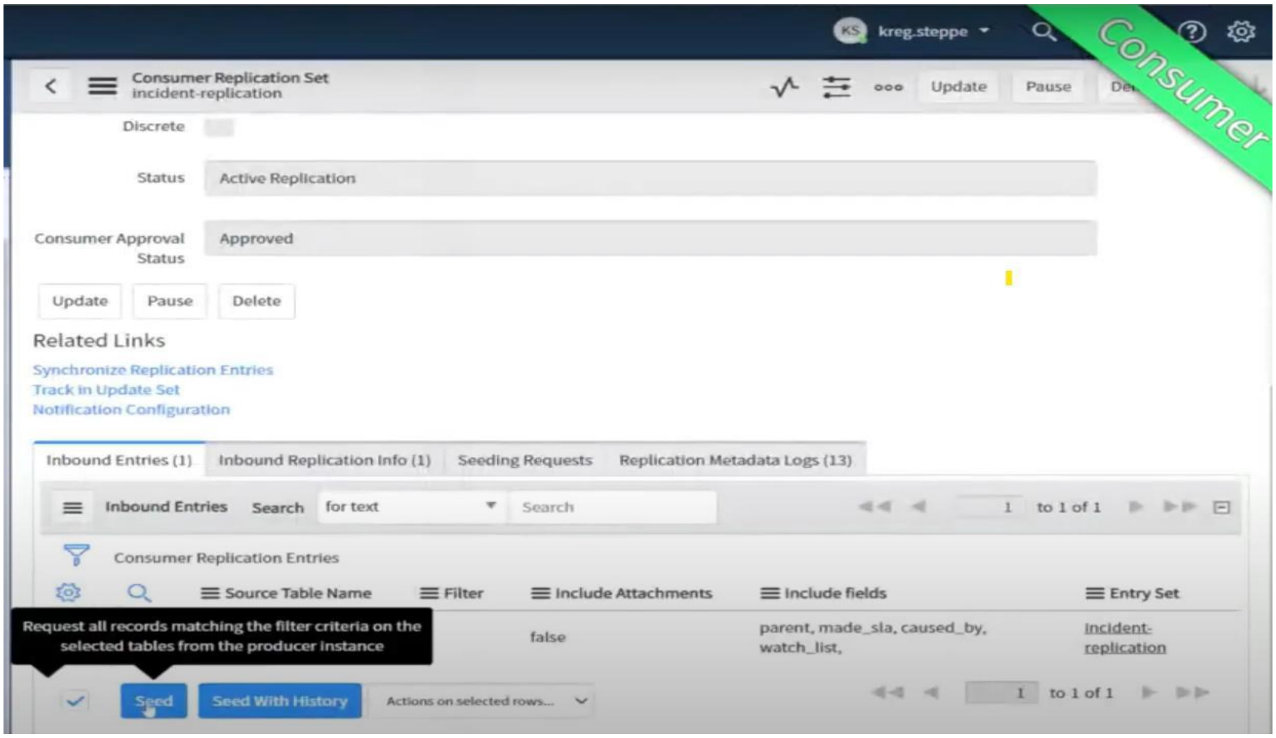
8. We can then activate the Consumer Replication Set



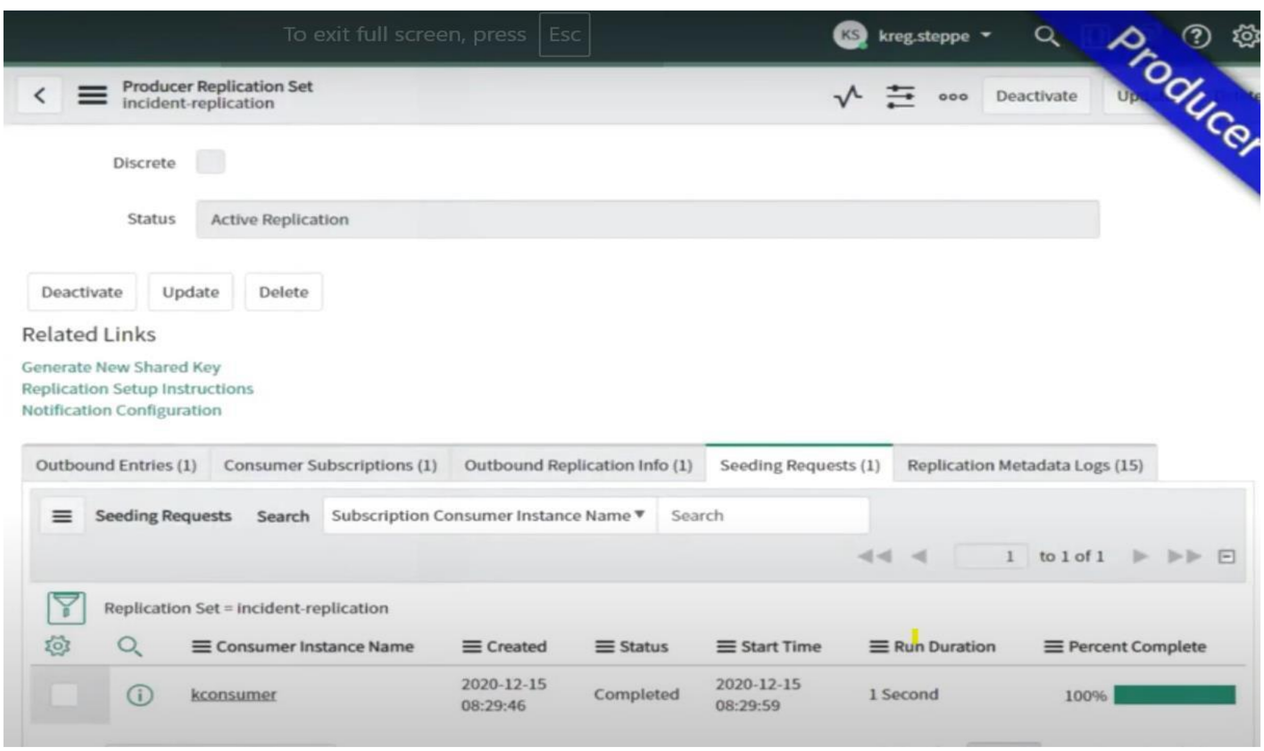
The status will get updated to **Active Replication**



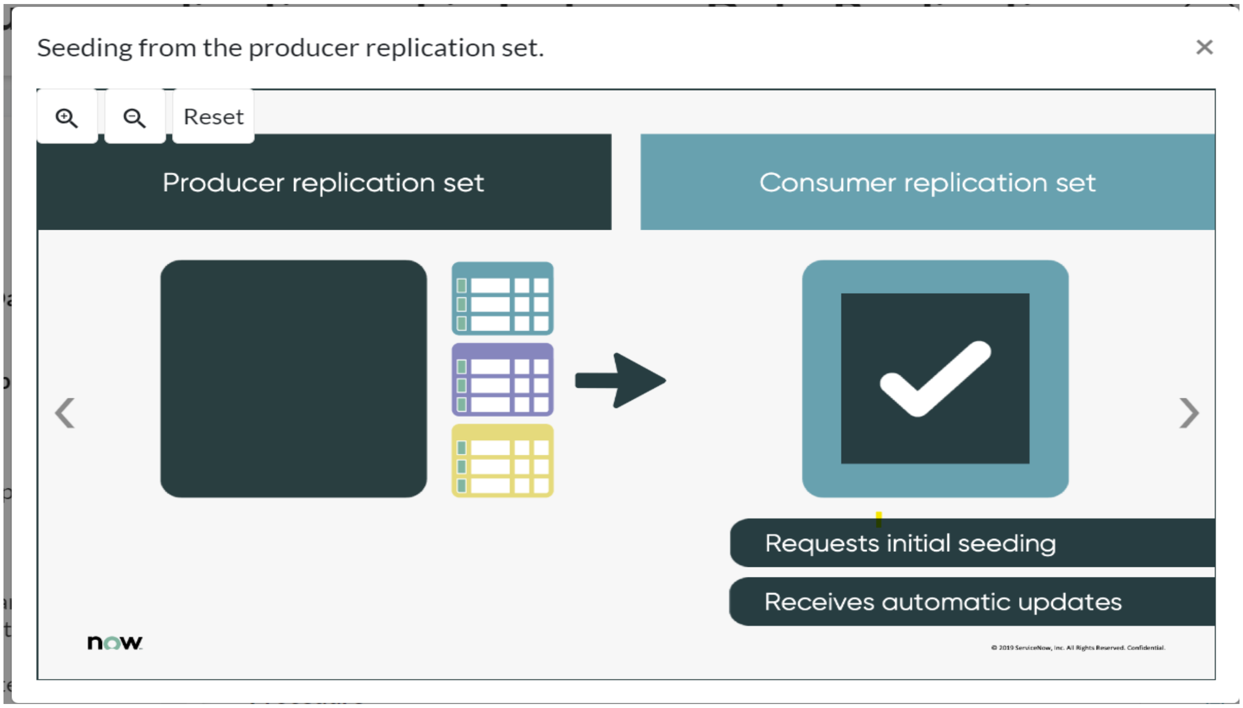
9. In the **Producer Replication Set,** we can **seed** the Table



Status of the Seeding Request



Seeding: Syncing the **producer** and **consumer** replication sets requires that you do a one-time download (called ***seeding***) of all the **producer** replication set data to the **consumer** instances.



You can initiate **seeding** requests on a **consumer** instance when you activate a **consumer** replication set. After **seeding**, replication involves data updates only. An audit trail contains a history of those record updates.

Note: Detailed information about **seeding** can be found here: https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/seed-consumer-instance.html

Transform replication data for Instance Data Replication

Used to Transform **producer** instance data for tables or table columns that are named differently on **consumer** instances in Instance Data Replication (IDR).

By default, the table data on a **producer** instance goes into the tables of the same name on **consumer** instances. **Transformation** is the process of replicating producer data in tables or table columns that have a different name on the **consumer** instances.

Note: Detailed information can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/task/modify-replicated-data.html

Adapters

Adapters modify the **producer** data before inserting the data on **consumer** instances in Instance Data Replication (IDR).

Adapters perform string and mathematical operations, such as converting one currency to another, or converting one time zone to another.

Each adapter has **Name** and **Description** fields. The name appears in the Adapter column. Use the **Description** field to explain the purpose of the data conversion.

Note: Detailed information can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/reference/adapter-descriptions.html

Steps to Revoke access to replicated data for Instance Data Replication

can be found here:https://docs.servicenow.com/bundle/xanadu-servicenow-platform/page/administer/instance-data-replication/task/revoke-consumer-sets.html

Monitoring Instance Data Replication

You can monitor the status of active producer and consumer replication sets, scheduled jobs, seeding requests, and license or usage details through the Instance Data

Replication (IDR) Monitoring Dashboard.

Accessing the IDR Monitoring dashboard

Users with the admin or idr_admin role can access the dashboard. Access the IDR Monitoring dashboard in the following way:

Navigate to Instance Data Replication > Monitoring Dashboard.

Choose either: IDR Overall Monitoring or IDR License and Usage Details dashboard.

The IDR Overall Monitoring dashboard monitors the following:

- Active Producer Replication Sets
- Active Consumer Replications Sets
- Suite of IDR Scheduled Jobs
- Producer Seeding Requests Within Last 7 Days
- Consumer Seeding Requests Within Last 7 Days

Note: Detailed information about the IDR Dashboard can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/concept/instance-data- replication-dashboard.html

Upgrading legacy replication sets to V2 in Instance Data Replication

1. Upgrade Instance Data Replication (IDR) legacy replication sets to V2 for improved message processing and replication performance. It utilizes **Hermes Messaging Service** for **Kafka** message transport. Offering **faster**, **scalable**, and **reliable** data replication

The Hermes Messaging Service is a Now Platform capability that's automatically enabled when the IDR plugin (com.glide.idr) is activated.

2. Upgrading replication sets from legacy to V2 is a seamless process. Active data replication from the producer to consumer continues throughout the upgrade, which

means you don't need to pause replication before you begin. Replication continues using the V2 replication sets after the upgrade is finished.

Note: Data sent from a V2 producer replication set is retained in the message queue for 24 hours.

Detailed Information about upgrading legacy replication sets to V2 in IDR can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/concept/upgrading-legacy- replication-sets-v2.html#upgrading-legacy-replication-sets-v2 section_r2w_4b5_lvb

Upgrade Eligibility:

- Both producer and consumer instances must be on **Utah** or higher.
- Some configurations may render certain legacy sets **not eligible**.

Note that some Utah legacy replication sets aren't eligible for upgrade to V2, depending on their configuration (for example, the replication set can't be

bidirectional or discrete)

- Check eligibility via the **Upgrade Eligibility** column in the Consumer Subscriptions list.

Compatibility:

If you upgrade a producer replication set to V2, you must upgrade the subscribed consumer replication sets to V2 as well to take advantage of the V2 processing improvements.

If you don't upgrade the consumers to V2, they continue to consume messages from the legacy producer replication set.

Upgrade Steps:

[https://docs.servicenow.com/bundle/vancouver-servicenow-](https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/upgrade-legacy-replication-sets.html)

[platform/page/administer/instance-data-replication/task/upgrade-legacy-replication-sets.html](https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/task/upgrade-legacy-replication-sets.html)

Revoke access to replicated data for Instance Data Replication

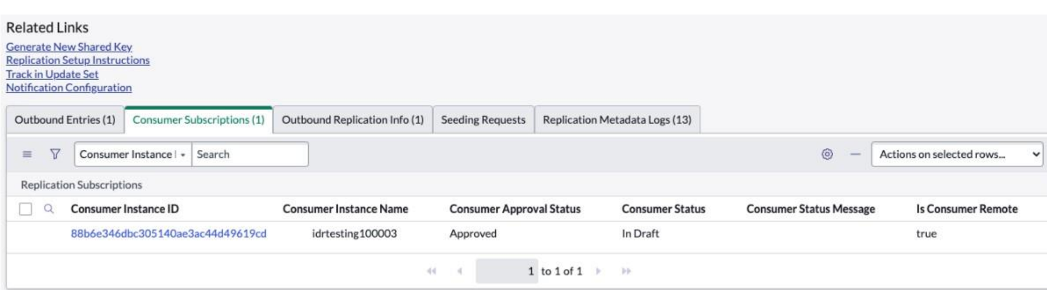
Revoke a consumer's access to replicated data if you believe that consumer instance should no longer receive data in Instance Data Replication (IDR).

When you generate a new, shared encryption key, approved consumer sets automatically receive the new key and data replication is not interrupted. Any consumer set with a status of Denied or Pending Approval does not receive the new encryption key, and replication ceases.

Procedure

1. On the producer instance, navigate to Instance Data Replication > Producer Replication Sets.
2. Select the producer replication set that has consumers whose access needs to be revoked.

The instance identifies the consumers under the Consumer Subscriptions tab.



3. On the Consumer Subscriptions tab, select the option for the consumer instance whose permissions you want to revoke.

4. In the Actions on selected rows list, select Revoke.

Instance Data Replication generates a new encryption key and shares it with all approved consumer sets. All other consumer sets stop receiving replication data. Admins of those consumer replication sets must reapply to restore access to the replication data.

IDR limitations and when not to use IDR

1. IDR runs in ServiceNow data centers only, which means there is no support for using IDR with on-premise instances.
2. Do not use IDR to clone instances.

IDR does not replicate metadata tables, child metadata tables, and most user and system tables. IDR is designed to replicate data, not clone instances.

3. Limitations for Replicating the updated Records:
   - Maximum size of records is 32MB
   - IDR supports replicating approximately 1 Million records per day

4. Limitation for Seeding Records:
   - Initial seeding of the tables must not exceed 3 million records per replication set.
   - Replication seeding must not take longer than 7 days to complete.
   - Producer and consumer instances must be in the same geographic region.
5. Avoid replicating system tables (tables with a sys_ prefix). If you replicate system tables (like sys_user, sys_user_group, or sys_user_grmember) that have existing data in the consumer instance, insert and update failures can occur during replication

For example the Application File [sys_metadata] table and tables that extend [sys_metadata] (including the Business Rules [sys_script], Catalog [sc_catalog], and Workflow [wf_workflow] tables) are excluded and can't be replicated.

6. Avoid replicating CMDB tables. If you determine the CMDB tables must be replicated, you must use conditions to constrain the count of replicated records and ensure all required columns are included in the replication set.
7. You can't replicate certain tables in Instance Data Replication (IDR). Child tables of tables in the exclusion list are also excluded.

Excluded tables

These tables are excluded from IDR:

- Tables with the prefixes V_PREFIX, SYSX_, sys, ts_, ua_, usageanalytics_, wf_, ecc_, clone_, jrobin_, rrd_, imp_, pf_, pfd_, idr_

sys_domain is an exception.

- Rotated tables, including syslog, sys_querystat, ecc_queue, ecc_event, cmdb_metric, sysevent
- Tables without a sys_id
- sso_federation
- saml2_update1_properties
- sso_properties
- digest_properties
- Instance

Child tables that extend these tables are also excluded from IDR and should not be included in replication sets because replication fails if they are included. For example, sc_cat_item extends sys_metadata, which is an excluded table. This makes sc_cat_item an excluded table due to inheritance.

Tables to avoid replicating:

- cmdb
- cmdb_ci

If you determine the CMDB tables must be replicated, you must use conditions to constrain the count of replicated records and ensure all required columns are included in the replication set.

8. Attachments

Avoid using HLA to replicate a series of large attachments on a regular basis. If you need to include attachments that are larger than 10 MB regularly, monitor HLA to ensure that the lag time doesn't exceed expectations.

9. You cannot replicate Edge Encrypted, column level encryption (CLE), and Password (2-Way Encrypted) fields.

Additional Information:

1. Information about monitoring data replication activities and troubleshoot potential issues in Instance Data Replication (HLA) can be found here:

https://docs.servicenow.com/bundle/vancouver-servicenow-

platform/page/administer/instance-data-replication/concept/administering-instance-data-replication.html

2. Information about resolving data replication errors in Instance Data Replication

https://docs.servicenow.com/bundle/vancouver-servicenow-platform/page/administer/instance-data-replication/reference/common-issues-idr.html